

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech (CE) Examination****November 2014****CE302/CE302.01 System Programming****Date: 17.11.2014, Monday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections and both must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Write entire option in case of multiple choice and matching questions.

SECTION – I**Q - 1 Do as Directed.****[A] Answer the following questions.****[06]**

- 1) Write RE (Regular Expression) for string starting with 10 and ending with 1 over the alphabet 0 and 1.
- 2) Write a difference between:
(a) Compiler and Interpreter (b) Token and Lexeme
- 3) What is symbol table? Which is the best data structure for implementation of symbol table?

[B] Match the following.**[05]**

- | | |
|--------------|---------------------------|
| 1) RDP | a) Use to define constant |
| 2) Reduction | b) Text editor |
| 3) #define | c) Top down parsing |
| 4) gedit | d) Parser |
| 5) RE | e) Bottom up parsing |
| | f) Scanner |

Q – 2 Answer the following questions. (Any three)**[12]**

- [A]** What do you mean by program relocation? Explain relocating loader in detail?
- [B]** What is nested macro? Explain in brief the use of stack in processing of nested macro.
- [C]** What is debugging? List and explain debugging facilities available in Turbo C++.
- [D]** What is the use of LTORG statement? What happens if LTORG statement is not used in program? Differentiate START and ORIGIN assembler directives.
- [E]** Write a short note on java interpreter.

Q – 3 Answer the following questions.

- [A] What do you mean by macro expansion? Explain two pass macro processor. [05]
- [B] What is the function of loader? Explain Bootstrap loader. [05]
- [C] What is a preprocessor? How does it differ from macro-processor? [02]

OR**Q – 3 Answer the following questions.**

- [A] How does a programmer decide whether to use a macro or a subroutine to accomplish a given logical functions? What are the similarities and differences between the use of macros at translation time and the use of subroutine at execution time? [05]
- [B] For the given program, do the following. [05]

```

      START    100
A      DS      5
M1     MOVER   AREG, B
      ADD     AREG, C
      MOVEM   AREG, D
D      EQU     A+1
M2     PRINT   D
      ORIGIN  A-1
C      DC      '5'
      ORIGIN  M2+1
      STOP
B      DC      '19'
      END

```

- 1) Show the contents of symbol table at the end of PASS-1 of two pass assembler.
- 2) Show the intermediate code generated for the program.

- [C] Which are the facilities provided by debugger? [02]

SECTION – II**Q – 4 Do as Directed.**

- [A] Choose correct or the best alternative in the following. [03]

- 1) Which analysis determines the meaning of a statement once its grammatical structure becomes known?
 (a) General (b) Syntax (c) Regular (d) Semantic

- 2) Symbolic names can be associated with _____.
 (a) Information (b) Operand (c) Data or instruction (d) mnemonic operation
- 3) Program generation activity aims at
 (a) Automatic generation of program
 (b) Skips generation of program
 (c) Organize execution of a program written in PL
 (d) Speeders generation of program

[B] State following statements are TRUE / FALSE.

[05]

- 1) Macro is a tiny little function.
- 2) Derivation is used in bottom up parsing.
- 3) A formal grammar is considered "context free" when its production rules can be applied regardless of the context of a nonterminal.
- 4) Predictive parsing does not support left recursion.
- 5) *Compiler* → *Assembler* → *Preprocessor* → *Linker* → *Loader* is a sequence of system software for processing a program.

[C] Define following in context of system programming.

[03]

- 1) System software 2) Semantic gap 3) Language Migrator

Q – 5 Answer the following questions. (Any three)

[12]

- [A]** Explain all data structures required in two pass assembler.
- [B]** Give details about the functions of editor.
- [C]** $A=B+C-D$ is the input to the compiler. What function will be performed by each phase? What will be the outcome of each of them? Describe using diagram.
- [D]** What is assembly language? List out the basic feature of assembly language. Also note down the advantages of assembly language.
- [E]** What is left factoring and left recursion? How to remove left factoring and left recursion from grammars? Explain with example.

Q - 6 Answer the following questions.

- [A] How forward referencing is handled in single pass assembler using TII (Table of Incomplete Instruction) and symbol table? Explain with example. [03]
- [B] Define FA (Finite Automata). List types of FA. Write a difference between them. [04]
- [C] What is a parse tree? What is an ambiguous grammar? Explain with example. [05]

OR

Q - 6 Answer the following questions.

- [A] Write a short note on language processor development tools. [03]
- [B] What are the functions of an assembler? List and explain the different type of assembly statements. [04]
- [C] Draw a DFA for $a(a|b)^*ba^*$ [05]
